Coroutine-Native I/O For C++29

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Coroutine-Native I/O For C++29

Abstract

This document presents a plan to standardize I/O for C++29. The plan is an eleven-paper series based on two libraries we built - Capy and Corosio - that implement coroutine-native I/O on C++20. Both are available today with independent adopters.

Revision History

R0: March 2026

- Initial version.

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1. Disclosure

We are the authors of Capy and Corosio, a complete I/O stack built on C++20 coroutines: timers, files, signals, sockets, TLS, DNS. We have a stake in the outcome.

Our design has boundaries. It cannot express compile-time work graphs. It does not support heterogeneous dispatch. It assumes cooperative multitasking. Those are real costs, and they define where coroutine-native I/O ends and where other models provide value.

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2. The Evidence

Every claim in this document is backed by published code. Every paper in the series is accompanied by a published library that implements it, a test suite, benchmarks where performance claims are made, and cross-platform CI. The committee can clone the repo and run the tests and examples.

2.1 What We Built

These two libraries represent the work to be standardized. Neither requires Boost:

Library	Role	Status
Capy	Abstract layer: execution model, buffer concepts, stream concepts, concurrency primitives	Published. Pure C++20. No platform dependency
Corosio	Platform layer: sockets, timers, DNS, TLS, signals	Published. IOCP, epoll, kqueue, select. Requires Capy

Software we are building on this foundation:

Library	Role	Status
Boost.Http	HTTP/1.1 server, sans-I/O	In Development. Compiled once against `any_stream`, ABI-stable
Boost.Burl	HTTP client	In Development

Library	Role	Status
Boost.Beast2	Successor to Boost.Beast	In Development
Boost.WebSocket	WebSocket protocol, sans-I/O	Planned

2.2 Independent Adopters

These libraries are maintained by other Boost authors. Each adoption is an independent data point:

Library	Author	Status
Boost.MySQL	Ruben Perez	Migrating to Corosio
Boost.Redis	Marcelo Zimbres Silva	Experimental port completed; conversion reports published on the Boost Developers Mailing List
Boost.Postgres	(upcoming)	Building on Corosio from day one

A production trading infrastructure company is exploring Corosio for high-performance networking. This is money on the wire.

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3. What We Found

Asio got many things right. We built on its stream model, its buffer sequences, its executor architecture. We started from C++20.

The committee designed C++20 coroutines for generality: async patterns, lazy evaluation, generators. We used them directly for I/O and found they resolve structural problems unique to C++ - problems that do not exist in Go, Rust, or Python, and that have stalled networking standardization for two decades.

Five properties of C++20 coroutines combine into a substrate suited to I/O:

1. **Type erasure is structural.** `coroutine_handle<>` erases the coroutine's type. The frame is the erasure. `task<T>` has one template parameter. `any_stream` works without per-operation allocation. The compiler provides the type erasure that template-based designs must build by hand.

2. **Promise customization.** `promise_type` controls allocation, suspension, resumption, and error handling - exactly the control points I/O needs. The IoAwaitable protocol uses `await_suspend` to propagate the executor, stop token, and frame allocator. No language extensions required.

3. **Stackless frames.** Each coroutine frame is independently allocated, independently suspendable, independently resumable. This maps directly to how operating systems do I/O: each operation suspends independently, the kernel completes them in any order, the reactor resumes them individually.

4. **Symmetric transfer.** `await_suspend` returning `coroutine_handle<>` enables zero-overhead context switching. If already on the correct thread, return the handle for direct resumption; otherwise post and return `noop_coroutine()`. No stack buildup.

5. **Frame as state.** Local variables survive across suspension in the compiler-generated frame. The coroutine frame is always allocated. The state it carries is free relative to the frame cost. This subsidizes type erasure: `any_stream` wraps any transport behind a vtable, and the operation state lives in the caller's frame, which already exists.

No single property is the insight. The five properties converge to resolve problems unique to C++:

Problem	Why C++-specific	Coroutine resolution
Template explosion	Go has no templates	`coroutine_handle<>` type erasure eliminates template metaprogramming
Compile times	Rust does not have this problem at C++ scale	`any_stream` compiles once and ships as a binary
Allocation control	Python has no user-facing allocator model	Frame allocator propagation gives users control where it matters
ABI stability	The committee's twenty-year wound	`any_stream` 's ABI is based on a forty year-old contract

The committee built five independent language mechanisms. Together they produce a design where correctness, ergonomics, and performance align rather than trade against each other.

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4. Design Criteria

We identified eight requirements for I/O in the standard. Here is how our design addresses each.

4.1 Zero overhead for features not used

The three-layer architecture (Section 6) gives every I/O object an abstract layer (virtual dispatch, ABI-stable), a concrete layer (protocol-specific, separately compilable), and a native layer (templated, fully inlined). The user who needs zero overhead uses the native layer. The user who needs ABI stability uses the abstract layer. Neither pays for the other.

4.2 Tools for building, not built-in solutions

Stream concepts are tools. Buffer concepts are vocabulary. The paper series does not propose an HTTP library or a WebSocket library. It proposes the abstractions that HTTP and WebSocket libraries are built from. The Boost libraries being ported or built on top of Corosio are evidence the model works.

4.3 ABI stability

`any_stream` has a fixed set of operations that it type-erases. Those operations will not change because the contract has not changed in forty years. Libraries that accept `any_stream&` compile once and ship as `.so` / `.dll` / `.a` files. New transports plug in without recompilation.

Coroutines make this possible. When a coroutine calls `co_await stream.read_some(buf)`, the caller's frame persists across suspension. That frame is already allocated. `any_stream` can type-erase for free because the operation state is not a template; it is independent of the coroutine that awaits it. The coroutine frame allocation we cannot avoid subsidizes the type erasure we want.

4.4 Backward compatibility

Pure C++20. No language extensions. No compiler intrinsics. Works with every conforming compiler today.

4.5 Production deployment

The libraries are available today. Cross-platform: IOCP, epoll, kqueue. Three independent adopters (MySQL, Redis, Postgres). One institutional deployment in production trading infrastructure. HTTP/1.1, HTTP/2, WebSocket, and TLS will be running on these abstractions.

4.6 Modularity

Each paper proposes the narrowest abstraction that delivers value on its own. Papers 1-4 have no platform dependency. Paper 2 (buffer concepts) has no async dependency. Each paper depends only on what came before in the series. No paper requires unfinished work by a different author.

4.7 Language-library co-design

The committee designed C++20 coroutines as language features. This series uses them as library infrastructure. The IoAwaitable protocol exploits `promise_type`, `await_suspend`, and symmetric transfer to propagate execution context, stop tokens, and frame allocators. Language design is library design.

4.8 Convergence with existing practice

Ecosystem	Read	Write	Buffer	Scatter/gather
BSD (1983)	`read()`	`write()`	`void*` + len	`readv` / `writev`
POSIX	`readv()`	`writev()`	`iovec`	`iovec[]`
Asio	`async_read_some()`	`async_write_some()`	`const_buffer`	`ConstBufferSequence`
Go	`io.Reader`	`io.Writer`	`[]byte`	`io.ReadFrom`

Ecosystem	Read	Write	Buffer	Scatter/gather
Rust	`AsyncRead`	`AsyncWrite`	`&[u8]`	`vectored`
.NET	`Stream.ReadAsync()`	`Stream.WriteAsync()`	`Memory<byte>`	`ReadOnlySequence<byte>`
Ours	`ReadStream::read_some()`	`WriteStream::write_some()`	`const_buffer`	`ConstBufferSequence`

Seven independent ecosystems. Same operations. Same shape. The standard is not inventing a new abstraction. It is formalizing forty years of convergence.

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5. Asio Continuity

This work builds on Asio, and we acknowledge that debt.

Asio has been used in production worldwide for over twenty years. It is the foundation of the Networking TS. Boost.Beast, Boost.MySQL, Boost.Redis, and hundreds of proprietary codebases depend on it. The committee has not been able to standardize it, but the operations it models - `async_read_some`, `async_write_some`, buffer sequences, executors - are the same operations every I/O framework arrives at independently.

The shift from C++11 to C++20 changes what is possible. Asio's named requirements become C++20 concepts. Its callback-based completion tokens become coroutine awaitables. Its stream model - the same `read_some` / `write_some` pair - gains structural type erasure through coroutine frames. The operations have not changed. The language has.

5.1 The Asio Adapter

Paper 1 delivers immediate value to existing Asio users. A small adapter wraps any Asio executor to satisfy the `Executor` concept. With this adapter, Asio users get `task<T>` as a drop-in replacement for `asio::awaitable<T>`.

Three gains:

1. **Physical insulation.** Today, `asio::awaitable<T>` forces the Asio dependency into every header that names the return type. `task<T>` confines the Asio dependency to the `.cpp` file that touches the socket. The header declares `task<response> handle_request(any_stream&)`. Consumers include neither Asio nor Corosio.

2. **Compilation improvement.** Fewer translation units include Asio headers. The heavy machinery - executor model, completion tokens, socket options - is confined to the files that need it.

3. **Backend insulation.** Swap Asio for Corosio by recompiling one `.cpp` file. Consumers never recompile. Headers do not change.

Frame allocator propagation also works through the adapter. Recycling allocators on MSVC yield a 3.1x throughput improvement (P4007R0[3] Section 5).

5.2 The Migration Path

Not "rewrite your application." Recompile one file.

The business logic accepts `any_stream&` and returns `task<T>`. The `.cpp` file that creates the socket includes Asio (or Corosio, or any conforming backend). The compilation boundary is the migration boundary. Cross it at your own pace.

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6. The Three-Layer Architecture

Every I/O object exposes three API layers.

Abstract. `io_stream`, `io_read_stream`, `io_write_stream`. Protocol-agnostic, virtual dispatch, separately compilable, no platform headers.

Concrete. `tcp_socket`, `timer`, `signal_set`. Full protocol-specific API, still virtual dispatch, still separately compilable.

Native. `native_tcp_socket<Backend>`. Templated on the platform backend. Member function shadowing eliminates the vtable. Full inlining.

The three layers share a single inheritance chain:

```
          io_stream                    abstract (Layer 3)
              |
         tcp_socket                    concrete (Layer 2)
              |
native_tcp_socket<Backend>             native   (Layer 1)
```

Property	Abstract	Concrete	Native
Compilation speed	Fastest	Fast	Slowest (platform headers + templates)
Separate compilation	Yes	Yes	No
Call overhead	Virtual dispatch	Virtual dispatch	None (direct / inlined)
API surface	Protocol-agnostic (bytes)	Full protocol-specific API	Same as concrete, fully inlined
Use case	Libraries, generic algorithms	Application code	Hot paths, benchmarks

The author chooses the layer. Users who need ABI stability stay at the abstract layer. Users who need zero-overhead dispatch drop to the native layer. Neither forces a choice on the other.

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7. Approach to Standardization

We are:

- Users first,
- Implementors second, and
- Proposers third.

Our approach to standardizing I/O is a layered series of papers: each builds on the last, each is backed by implementation experience, and each delivers value on its own.

The series evolves in two stages, mirroring the library split. The first four papers correspond to Capy; the remainder correspond to Corosio.

7.1 Stage One: Pure C++ Abstractions

#	Paper	Abstraction
1	Coroutines for I/O	Coroutine execution protocol, executor model, launch
2	I/O Buffer Ranges	Range-based buffer concepts for scatter/gather I/O
3	Stream Concepts	Coroutine stream concepts for async byte I/O
4	Coroutine Runtime	Thread pools, strands, structured concurrency

Pure C++20. No platform code. These abstractions enable sans-I/O protocols in the ecosystem: HTTP, WebSocket, TLS wrappers.

If all we get is Stage One, the standard has delivered the vocabulary for the entire async I/O ecosystem. External libraries implement portable, platform-specific I/O in terms of `std::io`. The ecosystem delivers the platform. The standard delivers the vocabulary.

7.2 Stage Two: Platform I/O

#	Paper	Abstraction
5	Timers	Async timer operations
6	Signals	Async signal handling
7	Files	Async file I/O
8	TCP	Sockets, acceptors, endpoints, IP addresses
9	DNS	Async name resolution
10	UDP	Datagram sockets

#	Paper	Abstraction
11	TLS	Transport security wrappers

Within Stage Two, the first three papers have nothing to do with networking. The committee can standardize seven papers before anyone says the word "socket."

7.3 Risk Separation

Previous approaches focused on getting sockets right before anything ships. We get the abstractions right and let sockets follow.

Stage One defines the concepts. Stage Two provides implementations. Between them, the ecosystem experiments. Multiple socket implementations can compete: Corosio's, an io_uring-native one, an IOCP-optimized one, a minimal embedded one. They all satisfy the same `Stream` concepts. The standard does not pick a winner prematurely.

The ABI stability of Stage One makes this safe. `any_stream` is the boundary. The socket behind it can be replaced. Business logic never recompiles.

Ship the concepts. Let the ecosystem discover the best implementations. Standardize what works.

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8. The Paper Series

8.1 Paper 1: Coroutines for I/O

P4003R0[2] published. Targeting first LEWG review at Brno (June 2026).

The IoAwaitable protocol, executor model, and `task<T>`. Depends on nothing. This is the foundation.

Key types. `Executor` concept (`dispatch`, `post`, `context`), `execution_context` with service registry and frame allocator ownership, `executor_ref` (two-pointer type-erased executor), `io_env` (bundles executor, stop token, frame allocator), `IoAwaitable` concept, `task<T>` (lazy coroutine task, one template parameter), `run()` and `run_async()`.

What coroutines provide. The IoAwaitable protocol solves frame allocator timing through forward propagation via TLS. The frame allocator is available before `operator new` executes. No language extensions required. 3.1x throughput improvement using recycling allocators on MSVC (P4007R0[3] Section 5).

Shipping status. Capy implements the full protocol. Corosio builds a complete networking stack on it. All shipping today on Windows, Linux, and macOS.

Standalone value. The Asio adapter (Section 5.1). Every Asio user gets `task<T>` with frame allocator propagation. Business logic drops the Asio header dependency. Compile times improve. The I/O backend becomes swappable behind a compilation boundary.

8.2 Paper 2: I/O Buffer Ranges

Buffer vocabulary types for scatter/gather I/O. Depends on nothing. No async dependency. No coroutines. No executor. Pure vocabulary.

Key types. `const_buffer` and `mutable_buffer` (byte-region vocabulary), `ConstBufferSequence` and `MutableBufferSequence` (range concepts for scatter/gather), `DynamicBuffer` (growable buffer with prepare/commit semantics), `buffer_copy`, `buffer_size`, `buffer_empty`.

Relationship to `std::ranges`. The concepts build on `std::ranges` - `ConstBufferSequence` is defined as `std::ranges::bidirectional_range<T>` with a value type constraint. But a single `const_buffer` is not a `std::ranges::range`. It represents a byte region, not an iterable sequence. `ranges::size` counts elements; `buffer_size` sums bytes. `ranges::drop` operates at element granularity; buffer slicing operates at byte granularity across element boundaries. We tried `std::ranges`. Here is what works and what does not. The extension is minimal.

Convergence. POSIX `iovec`, Windows `WSABUF`, Asio `const_buffer`, libuv `uv_buf_t`, Go `[]byte`, .NET `Memory<byte>`. Six ecosystems, same shape, none of them `span`.

Shipping status. Shipping in Capy. Used by every Corosio I/O operation.

Standalone value. Today, every C++ project that does I/O invents its own buffer types. Standard buffer concepts create shared vocabulary: a database driver that accepts `MutableBufferSequence` works with any I/O stack that speaks the same concepts. This paper advances independently of Paper 1 on a parallel track.

8.3 Paper 3: Stream Concepts

Coroutine stream concepts for async byte I/O. Depends on Papers 1-2.

This paper gives C++ ABI-stable I/O.

Two families of concepts. Caller-owned buffers: `ReadStream` (`read_some`), `WriteStream` (`write_some`), `Stream` (both), refined into `ReadSource` (complete reads) and `WriteSink` (complete writes, `write_eof()`). Callee-owned buffers: `BufferSource` (`pull` / `consume`) and `BufferSink` (`prepare` / `commit`). Zero-copy when needed. Simple path when not.

Type-erasing wrappers. `any_stream`, `any_read_stream`, `any_write_stream`, `any_read_source`, `any_write_sink`, `any_buffer_source`, `any_buffer_sink`. One pointer, one vtable, zero per-operation allocation. Boost.Http uses `any_stream` throughout - compiled once, linked against, works with any transport.

Convergence. BSD `read` / `write`, POSIX `readv` / `writev`, Asio `async_read_some` / `async_write_some`, Go `io.Reader` / `io.Writer`, Rust `AsyncRead` / `AsyncWrite`, .NET `Stream.ReadAsync` / `WriteAsync`. Every language arrived at the same pair of operations.

Synchronous streams for free. `co_await` checks `await_ready()` first. If `true`, no suspension. Memory buffers, test mocks, zlib decompressors, base64 decoders all satisfy `ReadStream` by returning immediately-ready awaitables. A pipeline of `tcp_socket` to `tls_stream` to `decompression_stream` to HTTP parser works regardless of which layers suspend. The algorithm does not know the difference.

Shipping status. Shipping in Capy. Corosio's `tcp_socket`, `tls_stream`, and `io_stream` all satisfy `Stream`.

Standalone value. Paper 3 completes the bridge to existing I/O ecosystems. After Papers 1-3, an `asio::ip::tcp::socket` becomes an `any_stream` through one adapter in one `.cpp` file. Business logic compiles against the standard. The Asio socket is behind the wrapper. 90% of a networking application's code compiles against `std::io`.

8.4 Paper 4: Coroutine Runtime

Thread pools, strands, and structured concurrency for coroutines. Depends on Paper 1.

Key types. `thread_pool` (multi-threaded execution context), `strand<Ex>` (serializes coroutine execution over any executor), `any_executor` (owning type-erased executor), `system_context` (process-wide singleton), `when_all(awaitables...)`, `when_any(awaitables...)`.

Shipping status. Shipping in Capy. `when_all` and `when_any` used in Corosio's `tcp_server` for concurrent accept loops.

Standalone value. Paper 4 completes Stage One. After Papers 1-4, a user can write a full concurrent networking application against the standard: `task<>` coroutines accepting `any_stream&`, standard `DynamicBuffer` for parsing, `thread_pool` for execution, `strand` for serialization, `when_all` / `when_any` for concurrency. 90% standard. 10% behind adapters. When Stage Two ships, the adapters become unnecessary.

8.5 Papers 5-7: Timers, Signals, Files

Paper 5: Timers. `timer` with `wait()`, `expires_at()`, `expires_after()`, `cancel()`. Uses `std::chrono::steady_clock`. The simplest kernel interaction that proves the IoAwaitable protocol works end-to-end with a real operating system. Shipping in Corosio. Cross-platform.

Paper 6: Signals. `signal_set` with `add()`, `remove()`, `wait()`, `cancel()`. Signals complete the server lifecycle: `co_await when_all(accept_loop(), signal_set.wait())` is graceful shutdown in one line. `<csignal>` is from 1989 and nearly unusable with modern C++. Shipping in Corosio.

Paper 7: Files. Async file I/O: `file` with `read_some()`, `write_some()`, `open()`, `close()`, and positioning. A `file` satisfies `Stream`. The same generic algorithms work on files and sockets. On Linux, `io_uring`. On Windows, IOCP with `FILE_FLAG_OVERLAPPED`. To be built in Corosio on solved infrastructure.

8.6 Papers 8-10: TCP, DNS, UDP

Paper 8: TCP. `tcp_socket`, `tcp_acceptor`, `endpoint`, `ipv4_address`, `ipv6_address`. A `tcp_socket` satisfies `Stream`. This delivers what the committee has been trying to standardize since N1925[5] (2005). Shipping in Corosio. Used by other Boost libraries and users.

Paper 9: DNS. `resolver` with `resolve(host, service)` for forward resolution and `resolve(endpoint)` for reverse resolution. Without DNS, TCP sockets can only connect to hardcoded IP addresses. Shipping in Corosio.

Paper 10: UDP. `udp_socket` with `send_to()`, `receive_from()`, `bind()`, and multicast support. UDP is the transport for DNS, QUIC/HTTP/3, game networking, media streaming, and IoT protocols. To be built in Corosio on solved infrastructure.

8.7 Paper 11: TLS

Transport security wrappers: `tls_context` for portable certificate management and `tls_stream` for encrypted I/O. A `tls_stream` wraps any `Stream` and satisfies `Stream` itself. The cryptographic engine is implementation-defined. ABI-stable by design.

The standard specifies the shape. The platform provides the encryption.

What the standard does. Portable certificate management (`tls_context`): certificates, trust anchors, verification modes, access to OS certificate store and revocation infrastructure. ABI-stable encrypted I/O (`tls_stream`): `handshake` , `read_some` , `write_some` , `shutdown` .

What the standard does not do. It does not reimplement OpenSSL. It does not mandate a specific engine. The cryptographic engine is implementation-defined - same as `std::filesystem` 's underlying OS calls. A TLS vulnerability disclosed on Monday is patchable by Tuesday. The fix does not wait for a standard library release cycle.

Convergence. Go `crypto/tls` , Rust `rustls` / `native-tls` , Python `ssl` , .NET `SslStream` , Java `SSLSocket` . Every language wraps TLS the same way: configuration object plus encrypted stream.

Shipping status. Shipping in Corosio with OpenSSL and WolfSSL backends. Both engines use the same abstract `tls_context` API.

The risk profile is lopsided: the implementation carries the risk; the interface carries none. The wrapper API has been stable since SSLv3 in 1996. Standardize the riskless part. Offload the risky implementation to the OS and ecosystem.

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9. `std::execution`

`std::execution` is an achievement. It serves its domains well.

9.1 What It Provides

Sender algorithms compose at compile time with full type information. The optimizer sees everything. Scheduler-based dispatch unifies CPU, GPU, and I/O scheduling under one vocabulary. The committee invested years in this work. It delivers real value for compile-time work graphs and heterogeneous dispatch.

9.2 Domain Separation

The question is whether byte-oriented I/O benefits from the same model.

The properties in Section 3 - type erasure, frame allocation, symmetric transfer - work because coroutines are used directly. A sender layer between the coroutine and the platform loses them. P4007R0[3] documents three structural gaps at the boundary where `std::execution` meets coroutines: error reporting, error returns, and frame allocator timing.

Domain specialization is not fragmentation. GPU compute got `nvexec` with CUDA extensions and a separate namespace. C++ has multiple container types, multiple string types, multiple smart pointer types. Two async models for two distinct domains is the same principle.

9.3 Convergence Points

The stream concepts impose no async-model dependency. A sender-based I/O library could define a `sender_stream` that satisfies `ReadStream` / `WriteStream` by returning sender-based awaitables. The concepts provide byte-oriented stream vocabulary that the entire ecosystem - including the sender ecosystem - can adopt.

The buffer concepts (Paper 2) have no async dependency at all. They are pure vocabulary for every I/O proposal.

9.4 Diversification Protects the Timeline

If the committee relies exclusively on one async path for networking and that path encounters delays, the networking timeline slips again. This has happened before.

Stage One costs the committee nothing to let proceed in parallel. Four papers of pure C++20 with no platform dependency. If sender-based networking succeeds, the buffer concepts and stream concepts enrich it. Nothing is wasted. If it encounters delays, the committee has an independent path ready.

SG14 published formal direction in the February 2026 mailing: "SG14 advise that Networking (SG4) should not be built on top of P2300R10[1]. The allocation patterns required by P2300R10 are incompatible with low-latency networking requirements. [...] Prioritize P4003R0[2] (or similar Direct Style concepts) as the C++29 Networking model." The domain experts whose requirements are the most stringent have independently concluded domain separation.

Letting both paths proceed is the conservative choice.

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10. Limitations

A coroutine-native I/O design has boundaries. Stating them plainly is part of the report.

- **Compile-time work graphs.** Coroutine-native I/O is a runtime model. It does not express static dataflow graphs where the compiler can reason about the entire pipeline at compile time. `std::execution` serves this domain.
- **Heterogeneous dispatch.** GPU compute, NUMA-aware scheduling, and cross-device dispatch require the scheduler abstraction that `std::execution` provides. Coroutine executors handle thread-pool and strand-based dispatch. They do not target hardware heterogeneity.
- **Cooperative scheduling.** The model assumes cooperative multitasking. Preemptive multitasking is outside scope.
- **Unbuilt papers.** Papers 7 (Files) and 10 (UDP) are not yet implemented in Corosio. The I/O context infrastructure already handles datagram descriptors and overlapped file handles. These wrappers are incremental work on solved infrastructure.

- **C library dependencies.** TLS wrappers depend on C libraries (OpenSSL, WolfSSL). The standard does not eliminate that dependency. It provides the stream abstraction that makes such wrappers composable and replaceable.

These boundaries define where coroutine-native I/O ends and where other models provide value. We draw the lines here so the committee does not have to.

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11. Timeline

Target: all eleven papers through LEWG by end of 2028. LWG wording through 2029H1.

11.1 Phase 1: Stage One Foundation (2026)

Quarter	Paper	Milestone
Q1 2026	Coroutines for I/O	P4003R0 published
Q2 2026	Coroutines for I/O	First LEWG review at Croydon
Q2 2026	I/O Buffer Ranges	Paper published
Q3 2026	Stream Concepts	Paper published
Q3 2026	I/O Buffer Ranges	LEWG review
Q4 2026	Coroutine Runtime	Paper published
Q4 2026	Stream Concepts	LEWG review

11.2 Phase 2: Stage Two Begins (2027 H1)

Quarter	Paper	Milestone
Q1 2027	Timers	Paper published
Q1 2027	Coroutine Runtime	LEWG review
Q2 2027	Signals, Files	Papers published
Q2 2027	Timers	LEWG review

11.3 Phase 3: Networking (2027 H2)

Quarter	Paper	Milestone
Q3 2027	TCP, DNS	Papers published

Quarter	Paper	Milestone
Q3 2027	Signals, Files	LEWG review
Q4 2027	UDP	Paper published
Q4 2027	TCP	LEWG review

11.4 Phase 4: Completion (2028)

Quarter	Paper	Milestone
Q1 2028	DNS, UDP	LEWG review
Q2 2028	TLS	Paper published
Q3 2028	TLS	LEWG review
Q4 2028	Full series	LWG wording review begins

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12. What We Continue to Maintain

We continue to maintain Capy and Corosio after standardization.

Production C++ lags the standard by three to seven years. A feature that lands in C++29 is not available to most production users until 2032-2036. Users who adopt Capy and Corosio today on C++20 need the Boost versions for years after the standard ships. This is the Boost.Filesystem pattern: precede the standard, coexist with it, persist for users on older compilers. Migration happens at the user's pace.

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13. `std::io`

```
std::io::task<>
handle_http_request( std::io::any_stream& stream )
{
    auto [ec, req] = co_await boost::http::read_request( stream );
    if( ! ec )
    {
        auto res = process_request( req );
        std::tie(ec) = co_await boost::http::write_response( stream, res );
    }
    co_return { ec };
}
```

We built this. It works. We are reporting what we found.

References

1. P2300R10 - " `std::execution` " (Michał Dominiak, Lewis Baker, Lee Howes, Kirk Shoop, Michael Garland, Eric Niebler, Bryce Adelstein Lelbach, 2024). https://wg21.link/p2300r10
2. P4003R0 - "Coroutines for I/O" (Vinnie Falco, Steve Gerbino, Michael Vandeberg, Mungo Gill, Mohammad Nejati, 2026). https://wg21.link/p4003r0
3. P4007R0 - "Senders and Coroutines" (Vinnie Falco, Mungo Gill, 2026). https://wg21.link/p4007r0
4. P4014R0 - "The Sender Sub-Language" (Vinnie Falco, 2026). https://wg21.link/p4014r0
5. N1925 - "A Proposal to Add Networking Utilities to the C++ Standard Library" (Chris Kohlhoff, 2005). https://wg21.link/n1925

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Chris Kohlhoff designed Asio's stream model, buffer sequences, and executor architecture. Twenty years of production deployment is the foundation this work builds on.

The authors of `std::execution` - Eric Niebler, Kirk Shoop, Lewis Baker, and their collaborators - invested years in a principled async framework for C++. Their work on sender/receiver serves domains where compile-time composition and heterogeneous dispatch matter. We respect that achievement and build alongside it.

The committee designed C++20 coroutines. Gor Nishanov, Lewis Baker, and their collaborators gave C++ the language mechanisms that make this I/O design possible.

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